

Member ID: _____

Time: _____

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COMPUTER NETWORK TECHNOLOGY

(300)

NATIONAL 2023

TOTAL POINTS (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What is the highest IP address that can be assigned to a host on the 192.168.10.0/24 network that uses the 255.255.255.0 subnet mask?
 - A. 192.168.1.254
 - B. 192.168.10.256
 - C. 192.168.10.254
 - D. 192.168.10.255
2. What is the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.224 subnet mask?
 - A. 14
 - B. 15
 - C. 16
 - D. 30
3. What is the advantage of using Jumbo frames?
 - A. More data is transferred in less packets
 - B. Improved network security
 - C. Faster Ping responses
 - D. All of the other answers are correct
4. Which protocol resolves IP addresses into MAC addresses?
 - A. DHCP
 - B. DNS
 - C. ARP
 - D. SNMP
5. Which of the following names is most used as the DNS prefix?
 - A. Server name
 - B. Domain name
 - C. NetBIOS name
 - D. Network adapter name
6. What is DHCP (Dynamic Host Configuration Protocol)?
 - A. DHCP is a network protocol used to authenticate user credentials
 - B. DHCP is a network protocol used to translate IP addresses to hostnames
 - C. DHCP is a network protocol used to statically assign IP addresses to network devices
 - D. DHCP is a network protocol used to assign IP addresses to network devices dynamically.

7. How many bits is an IPv4 address?
 - A. 32
 - B. 48
 - a. 64
 - b. 128
8. The file-sharing protocol, SMB, is primarily used by which of the following clients?
 - A. Microsoft
 - B. macOS
 - C. Android
 - D. Linux
9. How many bits is an IPv6 address?
 - A. 32
 - B. 48
 - C. 64
 - D. 128
10. What is MultiPath Input Output (MPIO)?
 - A. A third-party network performance application
 - B. A file sharing protocol
 - C. Multipathing used in iSCSI storage
 - D. Provides security updates to DHCP enabled clients
11. Which of the following DNS records would allow clients to resolve www.bpa.org and remote.bpa.org to the same IP address?
 - A. *.BPA.org
 - B. Any.BPA.org
 - C. Wildcard.BPA.org
 - D. \$_.BPA.org
12. Which of the following IP addresses are in the same subnet?
 - A. 10.0.0.10/8 and 100.0.0.10/8
 - B. 172.16.0.10/16 and 172.16.255.100/16
 - C. 192.168.10.10/24 and 192.168.100.100/24
 - D. 169.254.0.10/24 and 192.168.254.10/16
13. Which of the following is considered a bare metal Hypervisor?
 - A. VMware ESXi
 - B. Xen Server
 - C. Hyper-V
 - D. All of the other answers are correct

14. Which of the following is **true** in regard to containers and / or virtual machines?
- A. Containers have an operating system
 - B. Virtual machines do not have an operating system
 - C. Containers do not run on top of a hypervisor
 - D. Virtual machines run on top of a hypervisor
15. Which of the following is an open platform used to package and run applications in isolated environments, known as containers?
- A. Hyper-V
 - B. Docker
 - C. LXC
 - D. VMware
16. What is the purpose of having a quorum, with regards to a failover cluster?
- A. Provides a shared folder for transferring files between nodes
 - B. Prevents a split-brain scenario during a failover
 - C. Facilitates live migration between cluster nodes
 - D. None of the other answers are correct
17. Which of the following is known as an open-source container-orchestration system?
- A. Kubernetes
 - B. Microsoft Azure
 - C. Ansible
 - D. Amazon Web Services
18. What type of addresses are not used in IPv6?
- A. Unicast
 - B. Multicast
 - C. Anycast
 - D. Broadcast
19. What Protocol protects a layer 2 network from broadcast storms?
- A. Flow Control Protocol
 - B. Spanning Tree Protocol
 - C. Link Layer Discovery Protocol
 - D. Internet Control Message Protocol
20. What is one of the primary advantages of using a small form-factor pluggable (SFP)?
- A. Prevents ports from causing a loop
 - B. Allows bridging between multiple ports
 - C. Allows ports to be equipped with different types of transceivers
 - D. Prevents jumbo frames / packets from exceeding ~9000 bytes

21. Describe the TCP three-way handshake?
- A. ACK, SYN-ACK, and ACK
 - B. ACK-SYN, ACK, and SYN
 - C. SYN, SYN-ACK, and ACK
 - D. SYN, ACK, and SYN-ACK
22. Which of the following is the most redundant network topology?
- A. Ring
 - B. Matrix
 - C. Mesh
 - D. Star
23. What built-in protocol do you use to remote between Windows Server desktops?
- A. SIP
 - B. RTP
 - C. RDP
 - D. SNMP
24. What port does DNS use?
- A. 443
 - B. 389
 - C. 51
 - D. 53
25. The MAC address is associated with which layer of the OSI model?
- A. Layer 1 Physical
 - B. Layer 2 Data Link
 - C. Layer 3 Network
 - D. Layer 4 Transport
26. A new Server is added to the network with an IP address of 192.168.1.254/24 and can access the internet but cannot ping other local systems using hostnames. What is the issue?
- A. The default gateway is incorrect
 - B. The IP subnet is incorrect
 - C. There is an IP conflict
 - D. The DNS server is incorrect
27. Which IEEE Standard for Wi-Fi has a theoretical maximum throughput of over 9.6 Gbps?
- A. 802.11ac
 - B. 802.11n
 - C. 802.11g
 - D. 802.11ax

28. What is the correct sequence of message exchanges between a DHCP server and Client for a connection request?
- A. Offer, Discover, Acknowledge, Request
 - B. Discover, Offer, Request, Acknowledge
 - C. Discover, Request, Offer, Acknowledge
 - D. Discover, Acknowledge, Request, Offer
29. Which of the following accurately describes how account information is stored in a Domain?
- A. Account information is stored in a centralized database, on a server
 - B. Account information is stored in a local database, on each computer
 - C. Account information is stored on a shared folder, hosted on a file server
 - D. Account information is stored on a network device, such as a switch or router
30. Which of the following accurately describes how account information is stored in a Workgroup?
- A. Account information is stored in a centralized database, on a server
 - B. Account information is stored in a local database, on each computer
 - C. Account information is stored on a shared folder, hosted on a file server
 - D. Account information is stored on a network device, such as a switch or router
31. Which of the following statements is **true** when considering the primary differences between a proxy server and VPN?
- A. VPNs encrypt traffic
 - B. Proxy servers do not encrypt traffic
 - C. VPNs mask IP addresses
 - D. All of the other answers are correct
32. A ____ transmission send IP packets to a single recipient?
- A. Multicast
 - B. Unicast
 - C. Broadcast
 - D. Single mode
33. Users would like to access the company's private network while working from home and is unable to RDP into the File Server at the office. Which of the following would be a secure option?
- A. On the company gateway firewall, forward port 3389 to the server to allow external RDP connectivity
 - B. Connect the server directly to the ISP device, bypassing the router, so that it receives an external IP address
 - C. Allow users to share a login to third party, remote connectivity software
 - D. A VPN connection

34. You have a computer with an address of 169.254.32.23 and a subnet mask of 255.255.0.0. You cannot connect to your local file server IP 10.10.10.100. What is most likely the problem?
- A. The server cannot reach the DHCP server
 - B. The DNS servers are incorrect or are down
 - C. NetBIOS over TCP/IP has not been enabled
 - D. The Default Gateway is incorrect
35. A network administrator is troubleshooting an issue in which clients can ping local devices on the same network (192.168.1.0/24) but cannot reach the Internet. Which of the following is the most likely culprit?
- A. Router
 - B. Switch
 - C. Cables
 - D. Clients
36. Which of the following security access control methods uses the physical address of the NIC to limit access to a network?
- A. MAC filtering
 - B. ACL
 - C. VLAN
 - D. IP address blocking
37. Which of the following statements is **true** when considering the primary differences between TCP and UDP?
- A. TCP is a connectionless protocol
 - B. TCP is slower than UDP
 - C. UDP is a connection-oriented protocol
 - D. UDP is slower than TCP
38. What CLI command do you use to display active TCP or UDP connections?
- A. nbtstat
 - B. netstat
 - C. ipconfig
 - D. nslookup
39. What is a DHCP scope?
- A. A reserved range of IP addresses
 - B. A defined range of IP addresses that the DHCP server can assign to clients
 - C. A Dynamic Host Control Protection firewall rule
 - D. All of the other answers are correct.

40. What is a DNS server?
- A. A server gateway that provides access to the internet
 - B. A server that provides a database of file shares on the network
 - C. A server that contains a database of hostnames to IP addresses
 - D. A server that provides IP addresses to clients
41. Which of the following accurately describes DHCP snooping?
- A. Inspects DHCP requests and forwards them appropriately
 - B. Aggregates DHCP requests, so that multiple requests can be answered simultaneously
 - C. Automatically pushes security updates to DHCP enabled clients
 - D. Prevents unauthorized DHCP server access and drops unacceptable DHCP traffic
42. When implementing an iSCSI storage network, which of the following is something that would be considered beneficial?
- A. High latency
 - B. Multiple paths
 - C. Low bandwidth
 - D. All of the other answers are correct
43. A network administrator is configuring a storage network to use jumbo frames. Which of the following Windows ping commands, run from the server, could be used to ensure that jumbo frames are functional?
- A. ping <storageIP> -f -l 8972
 - B. ping <storageIP> -f -l 1500
 - C. ping <serverIP> -f -l 1400
 - D. ping <serverIP> -f -l 1500
44. Which port does a DHCP server use?
- A. 80
 - B. 67
 - C. 23
 - D. 69
45. What CLI command do you use to clear the DNS cache on a local system?
- A. ipconfig /registerdns
 - B. ipconfig /cleardns
 - C. ipconfig /flushdns
 - D. ipconfig /releasedns
46. What type of address is 169.254.2.255?
- A. Multicast address
 - B. Broadcast address
 - C. APIPA
 - D. Anycast address

47. Which layer of the OSI model is used to create a FTP connection to transfer files?
- A. Physical
 - B. Session
 - C. Application
 - D. Presentation
48. What command uses ICMP to test network connectivity?
- A. NSLOOKUP
 - B. IFCONFIG
 - C. PING
 - D. FTP
49. The default subnet mask for a Class C IP address is 255.255.0.0.
- A. True
 - B. False
50. What does the **arp -a** command on a Windows client show?
- A. IP address
 - B. MAC address
 - C. IP to MAC address mapping
 - D. None of the other answers are correct